

SET THEORY

Union, Intersection & Venn Diagrams

∈ Sets

∪ Union

∩ Intersection

■ Venn Diagrams

What is a Set?

A **set** is a collection of things or objects. We use **curly braces { }** and **capital letters** to write sets.

Examples of Sets

$A = \{\text{apple, banana, mango}\}$ — favourite fruits

$B = \{\text{football, cricket, badminton}\}$ — sports you play

$C = \{\text{red, blue, green}\}$ — colours

Elements of a Set

Each item inside the curly braces is called an **element**. We use the symbol $\in$ ("is an element of") to show membership.

$\text{football} \in A$ means "football is an element of A"

$\text{mango} \notin B$ means "mango is NOT an element of B"

Understanding Union ($A \cup B$)

The **union** of two sets is **all the elements** that are in A or in B or *in both*. Think of it as combining everything together — no duplicates!

Rule

$A \cup B$ = everything in A + everything in B (no repeats)

Symbol

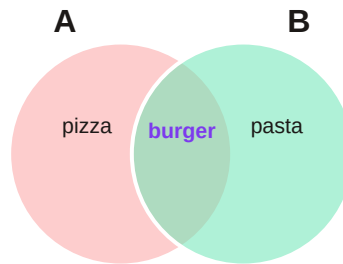
$\cup$ (looks like the letter U — think "Union"!)

Worked Example:

$A = \{\text{pizza, burger}\}$

$B = \{\text{burger, pasta}\}$

$$A \cup B = \{\text{pizza, burger, pasta}\} \leftarrow \text{burger appears once!}$$

Venn Diagram — Union (entire shaded area):

Understanding Intersection ($A \cap B$)

The **intersection** of two sets contains **only the elements common to both**. It's the overlap — what both sets share!

Rule

$A \cap B$ = only elements that appear in BOTH A and B

Symbol

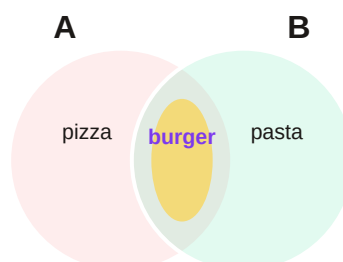
$\cap$ (looks like an upside-down $\cup$ — the **common part**!)

Worked Example:

$A = \{\text{pizza, burger}\}$

$B = \{\text{burger, pasta}\}$

$$A \cap B = \{\text{burger}\} \leftarrow \text{only burger is in BOTH sets!}$$

Venn Diagram — Intersection (overlapping area only):

What is a Venn Diagram?

A **Venn diagram** is a picture that uses overlapping circles to show relationships between sets. It makes it easy to see what is shared and what is unique.

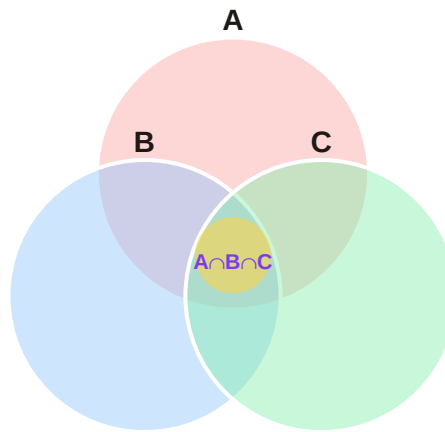
Step	What to Do
1	Identify each set's elements
2	Look at the overlapping area → that's the Intersection
3	Look at the whole shaded region → that's the Union
4	Label elements in the correct region
5	Practise with examples!

Three-Set Venn Diagrams (A, B, C)

When we have **three sets A, B, and C**, we use three overlapping circles. There are now seven distinct regions!

Region	Meaning
Only A / Only B / Only C	Outer parts — unique to each set
$A \cap B$, $B \cap C$, $A \cap C$	Where exactly two circles overlap
$A \cap B \cap C$	The very centre — shared by ALL three sets

Three-Set Venn Diagram:



Union: $A \cup B \cup C$ = all elements in any of the three sets

Intersection: $A \cap B \cap C$ = only elements found in ALL three sets

Example — Hobbies:

$A = \{\text{dancing, drawing, reading}\}$, $B = \{\text{singing, drawing, painting}\}$, $C = \{\text{drawing, reading, painting}\}$

$A \cup B \cup C = \{\text{dancing, singing, drawing, reading, painting}\}$

$A \cap B \cap C = \{\text{drawing}\}$ — the hobby everyone shares!

Real-Life Examples

School Subjects ■

$A = \{\text{Math lovers}\}$, $B = \{\text{Science lovers}\}$, $C = \{\text{English lovers}\}$

♦ $A \cap B \cap C$ = students who love ALL three subjects!

Food Favourites ■

$P = \{\text{pizza lovers}\}$, $I = \{\text{ice cream lovers}\}$, $C = \{\text{chocolate lovers}\}$

♦ $P \cap I \cap C$ = friends who love all three treats!

Sports Teams ■

$A = \{\text{Cricket, Football}\}$, $B = \{\text{Football, Badminton}\}$, $C = \{\text{Badminton, Cricket}\}$

♦ $A \cap B \cap C = \{\}$ — no sport is in ALL three (disjoint centre)!

Practice Problems

Problem 1 — Pets & Animals ■

$A = \{\text{cat, dog, rabbit}\}$

$B = \{\text{dog, fish}\}$

Questions:

1. $A \cup B = ?$ (Hint: List ALL elements from both sets)
2. $A \cap B = ?$ (Hint: Which element is in BOTH sets?)

Answers:

✓ $A \cup B = \{\text{cat, dog, rabbit, fish}\}$

✓ $A \cap B = \{\text{dog}\}$

Problem 2 — Colours ■

$A = \{\text{red, blue}\}$

$B = \{\text{green, yellow}\}$

Questions:

1. $A \cup B = ?$
2. $A \cap B = ?$ (Hint: Do these sets share any colours?)

Answers:

✓ $A \cup B = \{\text{red, blue, green, yellow}\}$

✓ $A \cap B = \{\}$ ← empty set! No common colours.

Problem 3 — Numbers ■

$A = \{2, 4, 6, 8, 10\}$ (even numbers ≤ 10)

$B = \{1, 2, 3, 4, 5\}$ (numbers 1–5)

Questions:

1. Find $A \cup B$

2. Find $A \cap B$

Answers:

✓ $A \cup B = \{1, 2, 3, 4, 5, 6, 8, 10\}$

✓ $A \cap B = \{2, 4\}$

Summary: What We Learned

- A **set** is a collection of objects; each item is an **element**.
- ⊃ **Union ($A \cup B$)** = all elements from both sets combined — no duplicates.
- ⊂ **Intersection ($A \cap B$)** = only the elements that appear in BOTH sets.
- **Venn diagrams** use overlapping circles to visualise set relationships.
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■ ■ With **three sets**, the centre region shows $A \cap B \cap C$.
- Sets help us organise, sort, and make decisions in real everyday life!

Sets are everywhere in life! ■